

Semi-prime factorization – Analysis and improvements on Fermat's Factorization method

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Foreword

This is a work in progress. I need to add a lot more information. However, in this iteration, I hope to compress everything down to as few pages as possible.

To do: Add references and closing thoughts and clean everything up.

Chapter I – Fermat's Factorization method

In its most basic form the procedure for Fermat's Factorization method is as follows (where N is the semi-prime being factored).

1. Calculate $\sqrt{N}$ in $\mathbb{Z}$
2. Starting from $x = \sqrt{N}$, calculate $f(x) = x^2 - N$ in a loop, incrementing x .
3. If $f(x)$ evaluates to a square, y^2 , calculate the greatest common divisor (GCD) on the difference of x and y .

Taking $N = 4387$ as an example:

Step 1 (Calculate the +/- square root of N):

$$\sqrt{4387} = 66 \text{ in } \mathbb{Z}$$

Step 2 (Starting from the square root calculate $f(x) = x^2 - N$:

$66^2 - 4387 = -31$ (not a square)
 $67^2 - 4387 = 102$ (not a square)
 $68^2 - 4387 = 237$ (not a square)
 $69^2 - 4387 = 374$ (not a square)
 $70^2 - 4387 = 513$ (not a square)
 $71^2 - 4387 = 654$ (not a square)
 $72^2 - 4387 = 797$ (not a square)
 $73^2 - 4387 = 942$ (not a square)
 $74^2 - 4387 = 1089 = 33^2$ (square)

Step 3 (If *the result is* also a square, calculate the GCD on the difference.):

$$74^2 = 33^2 \text{ mod } 4387$$

$$GCD(74 + 33, 4387) = 107$$

$$GCD(74 - 33, 4387) = 41$$

Both the Quadratic Sieve and Number Field Sieve algorithms (to do: insert references) are expansions on this method.

Chapter II – Analysis

a. Binomial expansion

As seen in chapter 1 the goal is the find the following numbers that we can use to take the GCD with:

$$GCD(x + y, N) = q$$

$$GCD(x - y, N) = p$$

x and y can thus be rewritten as the differences of the factors of N , $p + q$ or $p - q$.

In addition, many more such solutions exist created by $(ap + bq)/2$ and $(ap - bq)/2$ where a and b are multipliers.

If $a = 13$, $b = 5$ and $N = 4387$ (and thus factors $p = 41$ and $q = 107$)

We get:

$$(13 \times 41 + 5 \times 107)/2 = 534$$

$$(13 \times 41 - 5 \times 107)/2 = -1$$

And we can see this creates the following expression when we square 534 modulo N (4387 here):

$$534^2 - 4387 \times 13 \times 5 = 1^2$$

This behavior can be explained using binomial expansion.

I will shorten binomial expansion as $BE(x - y)^e$.

For example $BE(x - 74)^2 = x^2 - 148x + 5476$.

An interesting property of performing these binomial expansions is that we in essence simply increase the exponent on the binomial value.

For example:

$$BE(33 - 74)^1 = 33 - 74 = -41$$

$$BE(33 - 74)^2 = 33^2 - 148 \times 33 + 5476 = -41^2$$

$$BE(33 - 74)^3 = 33^3 - 222 \times 33^2 + 16428 \times 33 - 405224 = -41^3$$

$$BE(33 - 74)^4 = 33^4 - 296 \times 33^3 + 32856 \times 33^2 - 1620896 \times 33 + 29986576 = -41^4$$

Conversely we also see the following occurring if we substitute in the factor as root instead (or swapping a binomial term with the binomial value):

$$\begin{aligned} BE(41 - 74)^1 &= 41 - 74 = -33 \\ BE(41 - 74)^2 &= 41^2 - 148 \times 41 + 5476 = -33^2 \\ BE(41 - 74)^3 &= 41^3 - 222 \times 41^2 + 16428 \times 41 - 405224 = -33^3 \\ BE(41 - 74)^4 &= 41^4 - 296 \times 41^3 + 32856 \times 41^2 - 1620896 \times 41 + 29986576 = -33^4 \end{aligned}$$

In addition, if we have even exponents we can substitute the constant by a multiple of N and generate a polynomial p to solve for 0.

$$\begin{aligned} BE(41 - 74)^2 &= 41^2 - 148 \times 41 + 5476 \rightarrow p = 41^2 - 148 \times 41 + 4387 = 0 \\ BE(41 - 74)^4 &= 41^4 - 296 \times 41^3 + 32856 \times 41^2 - 1620896 \times 41 + 29986576 \rightarrow p = 41^4 - 296 \times 41^3 + 32856 \times 41^2 - 1620896 \times 41 + 4387 \times 6565 = 0 \end{aligned}$$

This behavior is rather crucial.

b. The discriminant

Of-course we cannot blindly perform binomial expansion on random numbers. We need some type of measure indicating us how close or far away we are from a solution.

Take for example the following binomial expansion $BE(x - 74)^2 = x^2 - 148x + 5476$ and on this quadratic polynomial p , we substitute the constant with N (for example $N = 4387$):

$$p = x^2 - 148x + 4387$$

If we now calculate the discriminant of p , $DISC(p) = 148^2 - 4 \times 4387 = 66^2$, we see that the result is also square. And we now have the square relation $148^2 = 66^2 \pmod{4387}$, whose $GCD(148 - 66, 4387) = 41$.

An observation:

From this we can make the following generalization that we can derive the factorization of a semi-prime N , if we find a quadratic polynomial with a constant set to N , or a multiple thereof, and whose discriminant yields a square. And from this we can also assume that the discriminant would yield a quadratic residue modulo any prime.

We also observe that the discriminant value is double the square of the binomial term. And can thus also be generated with $BE(x - 74)^2 = x^2 - 148x + 5476$ by substituting the root with a factor of N , as seen earlier. Since $BE(41 - 74)^2 = 41^2 - 148 \times 41 + 5476 = -33^2$. This is critical as calculating discriminant is not trivial as the polynomial degree goes up, but this gives us an alternative way to do it.

c. Using binomial expansion to create an oracle for the correct solution

The below examples assume $N = 4387$.

Previously we had the following binomial b :

$$b = 33 - 74 = -41$$

Next performing binomial expansion to degree 2, if we then substitute the constant by N to create polynomial p , and swap a binomial term with the binomial value (such that $b = 41 - 74$ instead), if p evaluates to 0, we have a valid solution.

$$BE(41 - 74)^2 = 41^2 - 148 \times 41 + 5476 = 33^2 \rightarrow p = 41^2 - 148 \times 41 + 4387 = 0$$

For binomial term $y = 74$, only binomial term $x = 33$ will yield a correct binomial value using this setup. Since 74^2 and 33^2 are congruent mod 4387. Further more calculating the discriminant on polynomial p will yield a square multiple of binomial term x , for example, $DISC(p) = (2 \times x)^2$ if we have a correct solution. See previous sub-chapter.

The main issue with this setup when performing binomial expansion using $BE(x - y)^e$, is that when expanding to even exponents, we need to know which multiple of N to substitute with. Because this will determine whether or not a root solution exists, derived from the binomial terms, that will evaluate the Quadratic to 0. Hence using this oracle setup we can restate the factorization problem as finding a multiple of N to substitute the constant value in $BE(x - y)^e$, where e is even, with. Luckily for us, we can calculate the possible multiples of N modulo m (where m is a product of primes). And this is what I propose as an improvement in the next chapter (Chapter 3 – Improvements).

d. An extended example

Let $N = 4387$

Let binomial $b = x - 534$.

$$\text{Hence } BE(x - 534)^2 = x^2 - 1068x + 534^2$$

We now need to substitute the constant in the resulting polynomial with a multiple of N . For example let multiple $k = 1$.

$$k = 1 \rightarrow p = x^2 - 1068x + 4387$$

$DISC(p)$ is not a square in $\mathbb{Z}$ hence we know $k = 1$ cannot be a valid solution.

However if $k = 65$:

$$k = 65 \rightarrow p = x^2 - 1068x + 4387 \times 65 \text{ and } DISC(p) = 2^2$$

The discriminant is defined as $DISC(p) = (2 \times x)^2$, where x is the binomial term, we can now substitute this into b .

$b(1) = 1 - 534 = -533$ and $p(533) = 0$. Since p evaluates to 0 in $\mathbb{Z}$ we know we have a correct solution and we can take the gcd using our binomial terms:

$$\begin{aligned} GCD(1 + 534, 4387) &= 107 \\ GCD(534 - 1, 4387) &= 41 \end{aligned}$$

In the next chapter we will see how to find these solutions in $\mathbb{Z}/p\mathbb{Z}$

Chapter 3 – Improvements

a. Calculating polynomial residues in $\mathbb{Z}/p\mathbb{Z}$

Let us restrict ourselves to binomial expansions to the second degree.

The way we calculate polynomial residues is that we must consider all possible coefficients in $\mathbb{Z}/p\mathbb{Z}$ whose discriminant evaluates to a quadratic residue.

Here we calculate all possible residues for the quadratic polynomial $ax^2 - bx + Nk$ where $N = 4387$ (you can also calculate them for $ax^2 + bx - Nk$ instead, it just changes the side we are calculating in the expression $x^2 = y^2 \bmod N$). We will ignore the cases where the leading coefficient is 0, as this would downgrade the degree of our polynomial. And in addition we will also ignore the case where $k = 0$ for the sake of brevity.

Mod 3

$k = 1, a = 1$ and $b = -1 \rightarrow x = \{2\}$, $k = 1, a = 1$ and $b = -2 \rightarrow x = \{1\}$, $k = 1, a = 2$ and $b = 0 \rightarrow x = \{1, 2\}$

$k = 2, a = 1$ and $b = 0 \rightarrow x = \{2, 1\}$, $k = 2, a = 2$ and $b = -1 \rightarrow x = \{1\}$, $k = 2, a = 2$ and $b = -2 \rightarrow x = \{2\}$

Mod 5

$k = 1, a = 1$ and $b = -2 \rightarrow x = \{4, 3\}$, $k = 1, a = 1$ and $b = -3 \rightarrow x = \{2, 1\}$, $k = 1, a = 2$ and $b = 0 \rightarrow x = \{3, 2\}$,
 $k = 1, a = 2$ and $b = -1 \rightarrow x = \{4\}$, $k = 1, a = 2$ and $b = -4 \rightarrow x = \{1\}$, $k = 1, a = 3$ and $b = 0 \rightarrow x = \{1, 4\}$,
 $k = 1, a = 3$ and $b = -2 \rightarrow x = \{2\}$, $k = 1, a = 3$ and $b = -3 \rightarrow x = \{3\}$, $k = 1, a = 4$ and $b = -1 \rightarrow x = \{3, 1\}$,
 $k = 1, a = 4$ and $b = -4 \rightarrow x = \{4, 2\}$

$k = 2, a = 1$ and $b = 0 \rightarrow x = \{4, 1\}$, $k = 2, a = 1$ and $b = -1 \rightarrow x = \{3\}$, $k = 2, a = 1$ and $b = -4 \rightarrow x = \{2\}$,
 $k = 2, a = 2$ and $b = -1 \rightarrow x = \{2, 1\}$, $k = 2, a = 2$ and $b = -4 \rightarrow x = \{3, 4\}$, $k = 2, a = 3$ and $b = -2 \rightarrow x = \{3, 1\}$,
 $k = 2, a = 3$ and $b = -3 \rightarrow x = \{4, 2\}$, $k = 2, a = 4$ and $b = 0 \rightarrow x = \{2, 3\}$, $k = 2, a = 4$ and $b = -2 \rightarrow x = \{4\}$,
 $k = 2, a = 4$ and $b = -3 \rightarrow x = \{1\}$

$k = 3, a = 1$ and $b = 0 \rightarrow x = \{3, 2\}$, $k = 3, a = 1$ and $b = -2 \rightarrow x = \{1\}$, $k = 3, a = 1$ and $b = -3 \rightarrow x = \{4\}$,
 $k = 3, a = 2$ and $b = -2 \rightarrow x = \{2, 4\}$, $k = 3, a = 2$ and $b = -3 \rightarrow x = \{1, 3\}$, $k = 3, a = 3$ and $b = -1 \rightarrow x = \{3, 4\}$,
 $k = 3, a = 3$ and $b = -4 \rightarrow x = \{2, 1\}$, $k = 3, a = 4$ and $b = 0 \rightarrow x = \{4, 1\}$, $k = 3, a = 4$ and $b = -1 \rightarrow x = \{2\}$,
 $k = 3, a = 4$ and $b = -4 \rightarrow x = \{3\}$

$k = 4, a = 1$ and $b = -1 \rightarrow x = \{2, 4\}$, $k = 4, a = 1$ and $b = -4 \rightarrow x = \{1, 3\}$, $k = 4, a = 2$ and $b = 0 \rightarrow x = \{4, 1\}$,
 $k = 4, a = 2$ and $b = -2 \rightarrow x = \{3\}$, $k = 4, a = 2$ and $b = -3 \rightarrow x = \{2\}$, $k = 4, a = 3$ and $b = 0 \rightarrow x = \{2, 3\}$,
 $k = 4, a = 3$ and $b = -1 \rightarrow x = \{1\}$, $k = 4, a = 3$ and $b = -4 \rightarrow x = \{4\}$, $k = 4, a = 4$ and $b = -2 \rightarrow x = \{1, 2\}$,
 $k = 4, a = 4$ and $b = -3 \rightarrow x = \{3, 4\}$

The problem is, that we can Chinese remainder these together to modulo 15, but then the solution set of possible residues grows quickly (the product of the size of both sets of solutions). So we need to come up with a smart way to generate solutions with this setup.

b. Singular quadratic roots vs non-singular quadratic roots

In the above residue mapping, we will observe that sometimes a set of coefficients will have a single root solution, and sometimes it will have two root solutions.

For example, looking in Mod 5 we see the following two coefficient, root pairings:

$k = 1, a = 4$ and $b = -1 \rightarrow x = \{3, 1\}$ Non-singular roots

$k = 1, a = 3$ and $b = -2 \rightarrow x = \{2\}$ Singular root

Singular roots indicate that the corresponding binomial term has prime p (5 here) as divisor and that the discriminant is also divided by prime p .

We have calculated the above residues using polynomials of the shape $ax^2 - bx + Nk$.

If we calculate the derivative in $\mathbb{Z}/5\mathbb{Z}$ we get the residue for linear coefficient b in $\mathbb{Z}/5\mathbb{Z}$ corresponding to a quadratic with the signs flipped $aax^2 + bx - Nk$ (the other side of the expression $x^2 = y^2 \bmod N$ that we are trying to solve for).

Calculating the derivative for coefficients with a singular root in $\mathbb{Z}/p\mathbb{Z}$ gives us a derivative that is 0 in $\mathbb{Z}/p\mathbb{Z}$. Since the the derivative calculates the linear coefficient for the other side of the expression $ax^2 - b_0x + Nk = ax^2 + b_1x - Nk$, and since the linear coefficient is constructed by multiplying the binomial term by 2, any divisors of the linear coefficient are hence also divisors of the binomial term.

For example using: $k = 1, a = 3$ and $b = -2 \rightarrow x = \{2\}$ we construct polynomial $p = 3 \times 2^2 - 2 \times 2 + 2$ in $\mathbb{Z}/5\mathbb{Z}$. And we see that the derivative $p' = 2 \times 3 \times 2 - 2 = 0 \bmod 5$ (or in other words 0 in $\mathbb{Z}/5\mathbb{Z}$).

This implies that the binomial term y , in binomial $b = y - x$ equals to 0 in $\mathbb{Z}/5\mathbb{Z}$.

Non-singular roots always yield non-zero residues for the binomial terms in $\mathbb{Z}/p\mathbb{Z}$.

C. p-adic lifting of polynomials in $\mathbb{Z}/p\mathbb{Z}$

Let us say we have the following coefficient and root pairing from our residue mapping:

$k = 4, a = 3$ and $b = -1 \rightarrow x = \{1\}$ in $\mathbb{Z}/5\mathbb{Z}$ gives us polynomial $p = 3 \times 1^2 - 1 \times 1 + 4387 \times 4 = 0 \bmod 5$

This gives us all of the following possible solutions in $\mathbb{Z}/25\mathbb{Z}$ where also the derivative yields 0 in $\mathbb{Z}/25\mathbb{Z}$, for the sake of brevity we will focus on these cases only:

$k = 4, a = 3$ and $b = -1 \rightarrow x = \{21\}$, $k = 4, a = 8$ and $b = -6 \rightarrow x = \{16\}$, $k = 4, a = 13$ and $b = -11 \rightarrow x = \{11\}$
 $k = 4, a = 18$ and $b = -16 \rightarrow x = \{6\}$, $k = 4, a = 23$ and $b = -21 \rightarrow x = \{1\}$

$k = 9, a = 3$ and $b = -11 \rightarrow x = \{6\}$, $k = 9, a = 8$ and $b = -16 \rightarrow x = \{1\}$, $k = 9, a = 13$ and $b = -21 \rightarrow x = \{21\}$

$k = 9, a = 18$ and $b = -1 \rightarrow x = \{16\}$, $k = 9, a = 23$ and $b = -6 \rightarrow x = \{11\}$

$k = 14, a = 3$ and $b = -21 \rightarrow x = \{16\}$, $k = 14, a = 8$ and $b = -1 \rightarrow x = \{11\}$, $k = 14, a = 13$ and $b = -6 \rightarrow x = \{6\}$
 $k = 14, a = 18$ and $b = -11 \rightarrow x = \{1\}$, $k = 14, a = 23$ and $b = -16 \rightarrow x = \{21\}$

$k = 19, a = 3$ and $b = -6 \rightarrow x = \{1\}$, $k = 19, a = 8$ and $b = -11 \rightarrow x = \{21\}$, $k = 19, a = 13$ and $b = -16 \rightarrow x = \{16\}$
 $k = 19, a = 18$ and $b = -21 \rightarrow x = \{11\}$, $k = 19, a = 23$ and $b = -1 \rightarrow x = \{6\}$

$k = 24, a = 3$ and $b = -16 \rightarrow x = \{11\}$, $k = 24, a = 8$ and $b = -21 \rightarrow x = \{6\}$, $k = 24, a = 13$ and $b = -1 \rightarrow x = \{1\}$
 $k = 24, a = 18$ and $b = -6 \rightarrow x = \{21\}$, $k = 24, a = 23$ and $b = -11 \rightarrow x = \{16\}$

Each of these solutions map to the same solution in $\mathbb{Z}/5\mathbb{Z}$. We see a total of 25 possible solutions in $\mathbb{Z}/25\mathbb{Z}$ generated from a single solution in $\mathbb{Z}/5\mathbb{Z}$. I refer to p-adic lifting, but usually it is used in the context of lifting roots of polynomials, but in the context that I use this term, it is also used to lift coefficient residues in $\mathbb{Z}/p\mathbb{Z}$ p-adically. So please keep that in mind. I tend to create these abstractions in my head and the names I give these abstract concepts may not be strictly correct in formal mathematical terms. But to me it would seem, that whether we are lifting root residues or coefficient residues, the underlying mechanism are very similar and connected.

D. Converting singular roots to non-singular roots and visa versa

Because we are searching for a two squares such that $x^2 = y^2 \pmod N$, and since both x and y can also be represented as binomials which can be expanded to the second degree, we can substitute the expression $x^2 = y^2 \pmod N$ with the more elaborate: $ax^2 - b_0x + Nk = ax^2 + b_1x - Nk$.

This also means that we can convert singular roots to non-singular roots, simply by transferring a root to the other side of the equation in $\mathbb{Z}/p\mathbb{Z}$. As long as either b_0 or b_1 does not have p as divisor.

For example $1x^2 + 66x - 4387 = 0 \pmod{37}$ has a singular root, while the converse $1x^2 - 0x + 4387 = 0 \pmod{37}$ has two roots.

$$b_1 = 66 \rightarrow 1 \times 4^2 + 66 \times 4 - 4387 = 0 \pmod{37}$$

$$b_0 = 0 \rightarrow 1 \times 4^2 - 0 \times 33 + 4387 = 0 \pmod{37}$$

$$b_0 = 0 \rightarrow 1 \times 33^2 - 0 \times 33 + 4387 = 0 \pmod{37}$$

But if we flip the sign on the linear coefficient b_1 , then the singular root becomes the other non-singular root for b_0 (we must also swap a and k , but both are 1 here):

$$b_1 = -66 \rightarrow 1 \times 33^2 - 66 \times 33 - 4387 = 0 \pmod{37}$$

This is a useful property. And note that $41 = 4 \pmod{37}$ and $107 = 33 \pmod{37}$, or in other words, residues for the factors of N . Hence this gives us the premise for our algorithm.

E. An algorithm outline

A most straight forward algorithm would be like this:

1. If we have the binomial $x - y$, we iterate values for y and create linear polynomials $yx - Nk = pv$, where N is the semi-prime being factored, k a multiplier and polynomial value pv is the B -smooth candidate over factor base B . And as x values, we sieve multiples of y . Such that we can easier control the factorization of yx too. As both the polynomial value and the value of the polynomial without the constant must factorize over B with this set-up.
2. Next perform binomial expansions $BE(x - y)^e$ and create polynomials of degree > 1 . And we use residue sieving to find factorization similar to the one found in step 1.

To do: Finish this.. not going to waste time here until PoC is ready though. Plus we can probably find a bunch of optimizations using p-adic lifting, so I want to implement that in code first.